

|          |                                                                             |           |
|----------|-----------------------------------------------------------------------------|-----------|
|          |                                                                             |           |
| 3(a)     | sum of potential energy and kinetic energy (of particles)                   | <b>B1</b> |
|          | (total) energy of random motion of particles                                | <b>B1</b> |
| 3(b)(i)  | no thermal energy <u>transferred</u>                                        | <b>B1</b> |
|          | work is done on the spring (increasing the potential energy of particles)   | <b>M1</b> |
|          | so internal energy increases                                                | <b>A1</b> |
| 3(b)(ii) | thermal energy transferred to water                                         | <b>B1</b> |
|          | work is done by water (expanding against atmosphere as it vaporises)        | <b>B1</b> |
|          | more thermal energy transferred than work done so internal energy increases | <b>B1</b> |